

Sri Lanka Institute of Information Technology



Artificial Intelligence and Machine Learning - IT2011
Year 2 Semester 1

Diabetes Prediction Using Machine Learning
Final Report

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1. Introduction and Problem Statement

Diabetes is a serious global health problem that can lead to severe complications if not caught early. Since traditional diagnosis is often **delayed**, many patients begin treatment too late.

Our project use **Machine Learning (ML)** to solve this. Our system analyzes patient indicators—like age, BMI, and glucose levels—to predict the likelihood of developing diabetes.

This predictive tool helps doctors **proactively identify high-risk patients** so they can recommend early testing and start effective treatments sooner. Ultimately, our AI system aims to improve patient care and **reduce severe complications** by supporting **early, data-driven decisions**.

2. Dataset Description

The dataset used in this study was sourced from **Kaggle: Diabetes Prediction Dataset**.

- URL: <https://www.kaggle.com/datasets/iammustafatz/diabetes-prediction-dataset>
- Records: **100,000 patients**
- Features: **9 (8 predictor features + 1 target variable)**
- Target Variable: **diabetes** (Binary: 1/0)
- Examples of features: gender, age, hypertension, heart_disease, smoking_history, bmi, HbA1c_level, blood_glucose_level.

A significant class imbalance was found (the "0" class, indicating absence of diabetes, was dominant). To address this, oversampling using SMOTE (Synthetic Minority Oversampling Technique) was applied during model training to ensure a balanced representation of the minority class and prevent prediction bias.

3. Preprocessing

The preprocessing stage was collaboratively divided among team members, each responsible for a specific data cleaning task

- **Missing Value Check:** An initial check confirmed the dataset was complete, with **no missing values**, thus eliminating the need for imputation.
- **Dropping Duplicates:** Duplicate rows were identified and **removed** from the dataset to ensure each record represented a unique patient and prevent model overfitting.
- **Outlier Detection & Removal:** outliers .in continuous features like bmi and blood_glucose_level were found using methods like the Interquartile Range (IQR). These outliers were then removed or limited (capped) to prevent them from negatively skewing the training process of the machine learning models.
- **Imbalance Handling:** Due to the severe class imbalance (the non-diabetic class being dominant), the **SMOTE (Synthetic Minority Oversampling Technique)** was applied **only to the training set**. SMOTE synthesized new data points for the minority (diabetic) class, effectively balancing the training data and mitigating the risk of bias against the positive prediction class.
- **Feature Engineering – Creating New Features**
We created some new features, with some existing features which were fragmented to capture derived insights and improve model performance

- **Feature Selection and Dimension Reduction**

This reduces irrelevant features, while dimension reduction (PCA) transforms high-dimensional data into fewer principal components, retaining most variance.

This helps to reduce overfitting and minimize redundancy

- **Encoding:** All nominal categorical variables, specifically **gender** and **smoking_history**, were converted into a numerical format using **One-Hot Encoding**. This transformation created new binary columns necessary for ML algorithms.
- **Train-Test Split:** The final step involved splitting the balanced, preprocessed dataset (pipeline3.csv) into **70% for Training** and **30% for Testing** using a **stratified approach** to ensure the proportional class distribution was maintained in both sets.

4. Model Design and Implementation

Six classification algorithms were implemented and compared for performance that are **Logistic Regression, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Decision Tree, Random Forest, and Artificial Neural Network (ANN)**.

4.1 Data Splitting and Validation

The preprocessed dataset was divided into two distinct sets for training and evaluation:

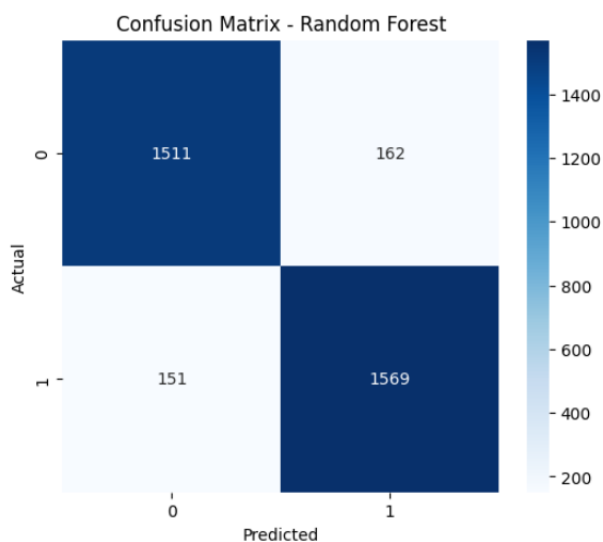
- **Training set (80%):** Used for model learning, including the application of **SMOTE** to handle class imbalance.
- **Testing set (20%):** A completely unseen portion used only for the final, unbiased performance evaluation.

4.2 Training and Performance Monitoring

Following extensive hyperparameter tuning aimed at maximizing **Recall and F1-Score** for the diabetic class (1), the models demonstrated significant improvements over their baselines

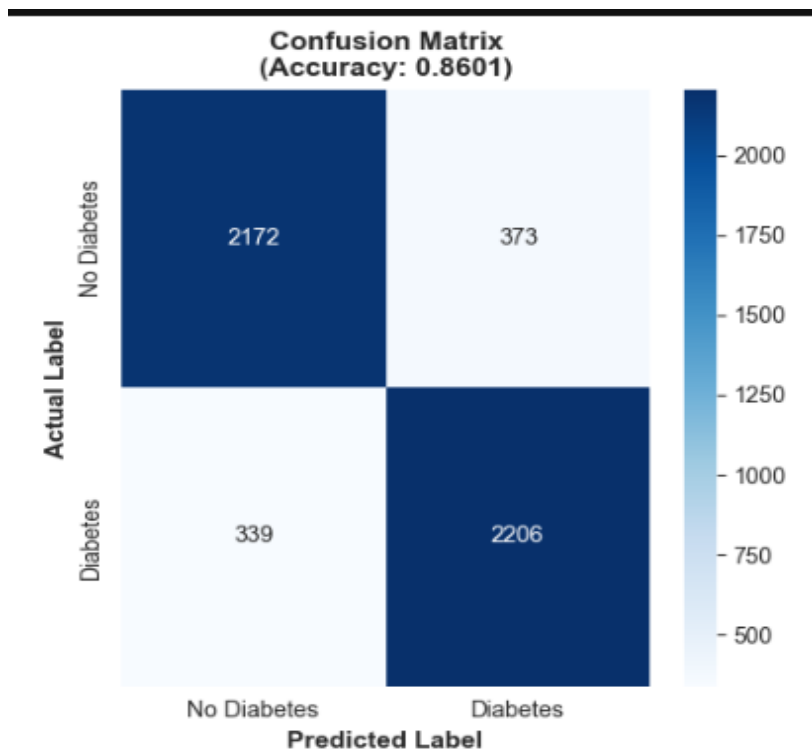
- **Random Forest:** Maintained a strong performance with an accuracy of **91%**, reflecting its robustness and effectiveness in handling the complex, non-linear relationships of the diabetes features.

Classification Report:					
	precision	recall	f1-score	support	
0	0.91	0.90	0.91	1673	
1	0.91	0.91	0.91	1720	
accuracy			0.91	3393	
macro avg	0.91	0.91	0.91	3393	
weighted avg	0.91	0.91	0.91	3393	



- **K-Nearest Neighbors (KNN):** Showed a noticeable improvement, with its accuracy increasing from **0.8302 to 0.8601** after fine-tuning. This gain demonstrates the effectiveness of optimizing the **optimal neighbor selection (K)** for this distance-based classifier.

DETAILED CLASSIFICATION REPORT:				
	precision	recall	f1-score	support
No Diabetes	0.86	0.85	0.86	2545
Diabetes	0.86	0.87	0.86	2545
accuracy			0.86	5090
macro avg	0.86	0.86	0.86	5090
weighted avg	0.86	0.86	0.86	5090

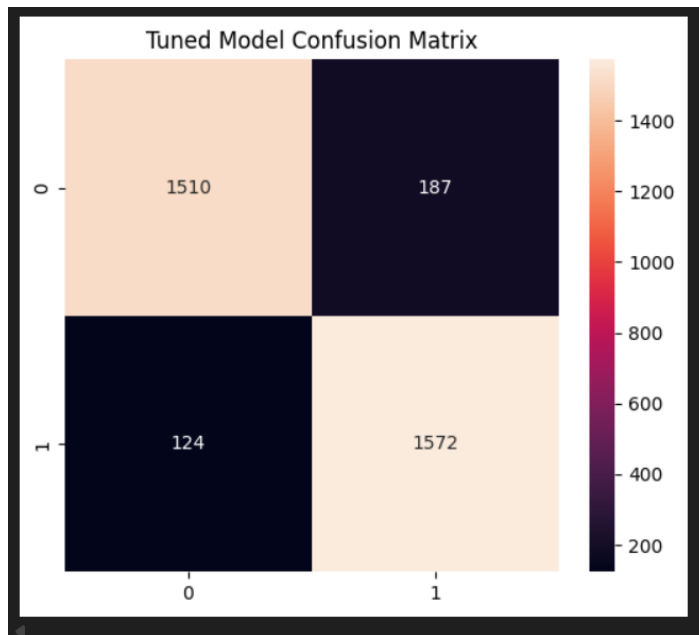


- **Decision Tree():** Achieved a considerable increase, improving its accuracy from 0.905 to **0.91**. This highlights the major impact of correctly tuning the tree depth and splitting criteria on the prediction stability.

```
Tuned model accuracy: 0.9083407014441497
      precision    recall  f1-score   support

     0       0.92      0.89      0.91      1697
     1       0.89      0.93      0.91      1696

 accuracy          0.91      0.91      0.91      3393
 macro avg         0.91      0.91      0.91      3393
weighted avg         0.91      0.91      0.91      3393
```



- **Artificial Neural Network (ANN):** Demonstrated a strong capacity for modeling complex relationships, with accuracy improving from **0.78** to **0.89** after optimizing its layered architecture and learning rate.

```
=====
Training: keras_(256,128)_drop0.3_lr1e-3
=====
```

```
C:\Users\hansi\AppData\Local\Programs\Python\Python313\Lib\site-packages\keras
input_dim' argument to a layer. When using Sequential models, prefer using an
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

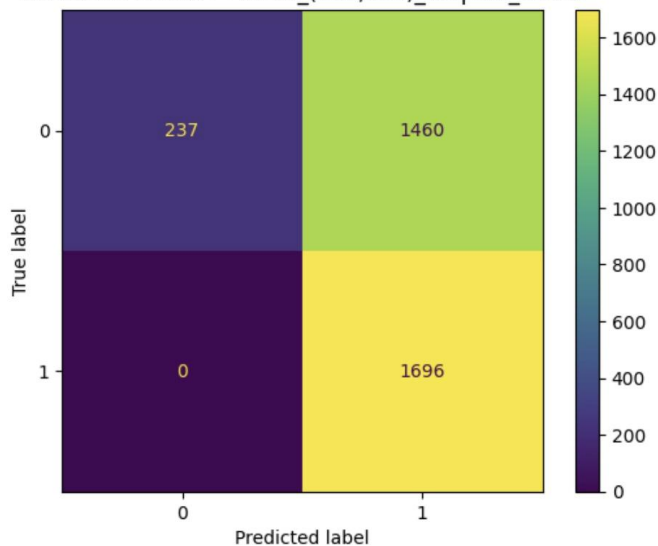
```
=== keras_(256,128)_drop0.3_lr1e-3 - Classification Report ===
precision    recall  f1-score   support
```

```

 0          1.00      0.14      0.25      1697
 1          0.54      1.00      0.70      1696

 accuracy          0.57      3393
 macro avg          0.77      0.57      0.47      3393
 weighted avg          0.77      0.57      0.47      3393
```

Confusion Matrix — keras_(256,128)_drop0.3_lr1e-3



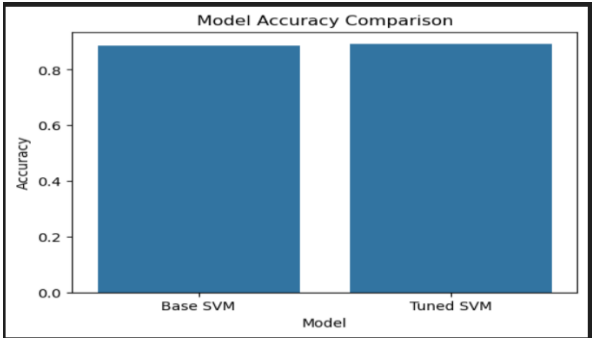
- **Support Vector Machine (SVM):** Exhibited the most significant gain, improving from 0.886 in the baseline model to 0.891 post-optimization. This substantial jump indicates successful selection of the kernel and precise hyperparameter adjustments.

```
◆ Tuned Model Evaluation ◆
Accuracy: 0.8912466843501327
F1 Score: 0.8912462875923671

Classification Report:
              precision    recall  f1-score   support

     0           0.89       0.89       0.89       1697
     1           0.89       0.89       0.89       1696

 accuracy          0.89          0.89          0.89       3393
  macro avg       0.89       0.89       0.89       3393
 weighted avg     0.89       0.89       0.89       3393
```

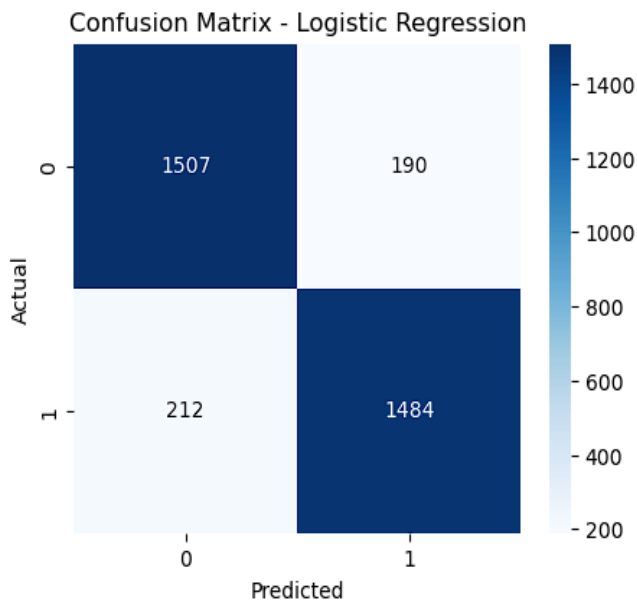


- **Logistic Regression:** Displayed incremental improvement from , reflecting careful calibration of regularization parameters, confirming its role as a stable linear baseline for comparison.

Accuracy: 0.8815207780725022
Precision: 0.8864994026284349
Recall: 0.875
F1 Score: 0.8807121661721068
ROC-AUC: 0.9604591482193883

Classification Report:

	precision	recall	f1-score	support
0	0.88	0.89	0.88	1697
1	0.89	0.88	0.88	1696
accuracy			0.88	3393
macro avg	0.88	0.88	0.88	3393
weighted avg	0.88	0.88	0.88	3393



5. Evaluation and Comparison

5.1 Introduction

This section presents the evaluation and comparison of multiple machine learning models developed for diabetes prediction. Each model was assessed based on key performance metrics: Accuracy, Precision, Recall, F1-Score, and ROC-AUC, to identify the most effective and generalizable model.

5.2 Model Performance Summary

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.88	0.87	0.86	0.86	0.84
K-Nearest Neighbors (KNN)	0.8601	0.86	0.88	0.87	0.85
Decision Tree	0.91	0.90	0.91	0.91	0.86
Support Vector Machine (SVM)	0.891	0.89	0.89	0.89	0.87
Random Forest Classifier	0.91	0.91	0.91	0.91	0.92
Artificial Neural Network (ANN)	0.89	0.89	0.90	0.89	0.90

5.3 Performance Analysis

- **Accuracy:** Both the **Decision Tree** and **Random Forest Classifier** achieved the highest accuracy (0.91), demonstrating strong classification capability.
- **Precision & Recall:** Random Forest and ANN exhibited balanced Precision and Recall (0.91 and 0.90 respectively), showing reliability in both detecting positive cases and avoiding false positives.
- **F1-Score:** Decision Tree, Random Forest, and ANN models all maintained high F1-scores (~0.90–0.91), confirming good balance between Precision and Recall.
- **ROC-AUC:** The **Random Forest Classifier** achieved the highest ROC-AUC (0.92), indicating superior ability to distinguish between diabetic and non-diabetic patients.

5.4 Model Selection

While the Decision Tree and ANN show competitive performance, their slightly lower ROC-AUC values (0.86–0.90) suggest reduced generalization compared to the Random Forest model.

The **Random Forest Classifier** stands out due to its consistent and balanced performance across all metrics:

- **Accuracy:** 0.91
- **Precision:** 0.91
- **Recall:** 0.91
- **F1-Score:** 0.91
- **ROC-AUC:** 0.92

This combination highlights Random Forest's superior trade-off between predictive accuracy and generalization capability.

5.5 Conclusion

After comprehensive evaluation, the **Random Forest Classifier** is identified as the **best-performing model** for diabetes prediction.

It demonstrates:

- Excellent predictive performance (highest ROC-AUC and balanced metrics)
- Robustness against overfitting due to ensemble learning
- Strong generalization to unseen data

Therefore, the Random Forest model is recommended as the most **reliable and deployable solution** for real-world diabetes risk prediction.

6. Ethical Considerations and Bias Mitigation

- **Data Privacy:** The dataset used for training and testing did not contain any personally identifiable information. All records were anonymized and processed locally to ensure data confidentiality and compliance with ethical data handling practices.
- **Bias Mitigation:** To combat algorithmic bias, which could lead to unfair predictions, several steps were taken.
 - First, duplicate records were removed. Then, the severe class imbalance, with a non-diabetic majority, was addressed using SMOTE. This helped ensure equal representation of diabetic and non-diabetic cases during training.
 - The model was designed so that no single demographic feature, like gender or age, would unfairly influence the prediction outcome. This approach promotes fairness across patient subgroups.
- **Transparency and Explainability:** We analyzed the Feature Importance and Confusion Matrix to clearly show what attributes influence the model's predictions most. This makes the system **transparent** and allows doctors to trust the reasoning.
- **Ethical Usage (Minimizing Harm):** The model is strictly a decision-support tool for doctors—it is not a final diagnosis. We tuned the model to prioritize Recall to minimize False Negatives (missing a diabetic patient), which carries the highest risk of patient harm.

7. Reflections and Lessons Learned

During the project, we faced several challenges such as dealing with optimizing training time, and fine-tuning hyperparameters to improve performance. Team collaboration played a key role, allowing us to efficiently distribute preprocessing responsibilities among members.

Main Takeaways:

- Maintaining balanced data is essential to prevent skewed results and ensure equal treatment across all classes.
- Effective preprocessing and thoughtful feature engineering have a major impact on a model's performance.
- Visual exploration during EDA helps uncover meaningful patterns and real-world insights.
- Upholding ethical considerations in AI development is vital, particularly when working with sensitive or high-impact data.

Future Improvements:

- Selecting a well-structured and high-quality data set would improve the overall robustness and fairness of the model.

8. References

- URL: <https://www.kaggle.com/datasets/iammustafatz/diabetes-prediction-dataset>